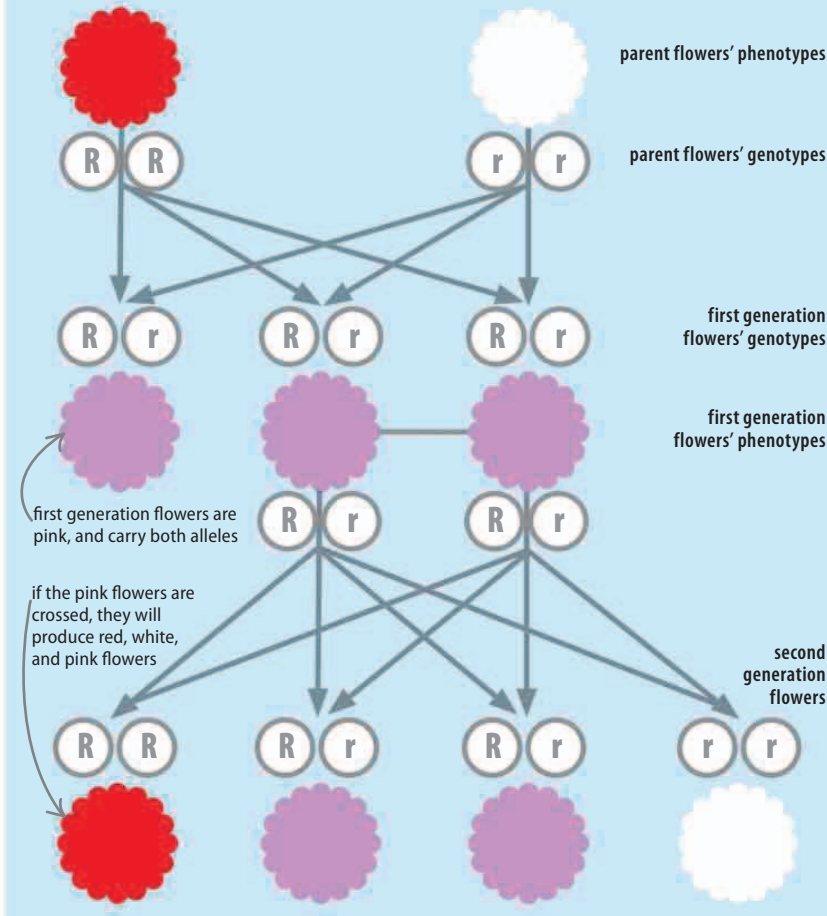


Codominance

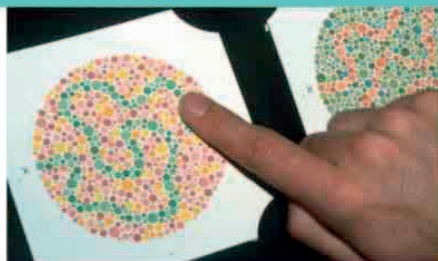
Not all alleles are dominant or recessive. Sometimes, both alleles are expressed at once in a system called codominance, or incomplete dominance. In the example below, the red parent flower has two red alleles (R), while the white parent flower has two white alleles (r). When the pair breed, all the offspring have the genotype Rr and codominance makes all the flowers pink. However, breeding two pink flowers produces red and white, as well as pink, blooms.



REAL WORLD

X-linked diseases

Males are more likely to be color blind, because they have a defective gene on their X chromosome and their shorter Y has no alternative allele for the problem. Females can carry the same defective gene, but have normal vision thanks to a healthy allele from their other X chromosome.

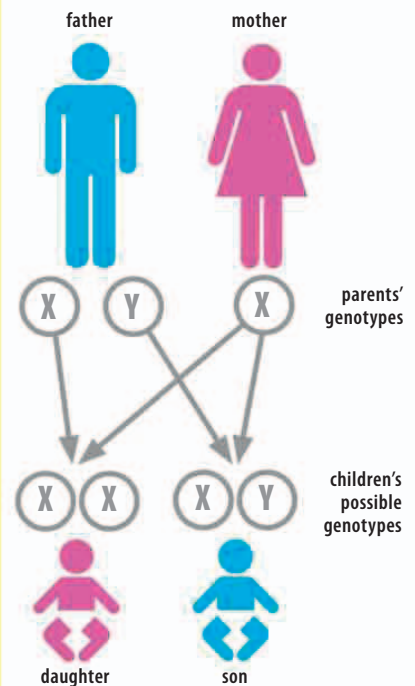


Sex chromosomes

A person's sex is determined by inheriting particular chromosomes from the mother and father. Females have two X chromosomes, while males have an X and a Y chromosome—the Y is much smaller, and has fewer genes, than the X. A mother's gametes always contain an X, while a sperm can have an X or Y.

▽ Determining gender

Because the mother always gives an X chromosome, it is the father's gamete that determines the sex of the baby. If an X sperm fertilizes the egg, the baby will be female, and male if a Y sperm achieves it.



Humans have **46 chromosomes**, which is less than some rats, at **92**, but more than kangaroos, which have only **16**.